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# Multi-filter dynamic graph convolutional networks for skeleton-based action recognition

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**Abstract**

Action recognition plays an important role in video understanding, and the dynamic changes of human bones provide important information for it. In order to solve the problem that the source point can only obtain the characteristics of adjacent nodes and the static structure of the skeleton diagram can not meet the adaptive requirements, this paper proposes an Inception structure and dynamic skeleton diagram based on the convolutional neural network, namely, the multi-filter dynamic graph convolutional neural network. The method can not only enable the source point to obtain the characteristics of distant nodes, but also can use the dynamic skeleton diagram structure to generate different body associations for different actions, which improves the adaptability and accuracy of the model and achieves better results.

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*Keywords:* Convolutional neural network; action recognition; skeleton diagram.

## 1. Introduction

In recent years, the research of action recognition is beginning to attract people's attention, due to the widespread use of network streaming media, it becomes urgent to analyze illegal actions in network video stream. For the detection of scenes such as violence and eroticism in videos, action recognition plays a very important role. In the past years of the research of action recognition, both methods based on RGB video and skeleton sequence have been

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studied a lot. The recognition method based on RGB video[6-7] mainly focuses on the modeling of optical flow. Since 2017, more and more research on behavior recognition based on skeleton sequence has been conducted, and the results obtained are also getting better and better. This is mainly due to two reasons, on one hand, with the development of the pose estimation algorithm, make the human body skeleton in the extraction of video features become easy to implement. On the other hand, the joints in the human body skeleton structure convey important information, than RGB video in the form of a more robust, so researchers begin to pay close attention to the gesture recognition algorithm based on skeleton sequence.

For action recognition of skeleton sequence, domestic and foreign scholars have carried out studies in the direction of convolutional neural network[10-11] and convolutional neural network[1-4] . Among them, based on the study of graph convolutional neural network, literature [1] proposed spatio temporal graph Convolutional Networks(ST-GCN), and applied the spatio temporal graph model to human action recognition. On this basis, literature [2] divided the human body structure into parts, and improved the recognition effect by extracting the features of different parts. literature [3] takes the nodes and edges of skeleton graph as the input model of a two-stream network, and also achieves good results. Although the above action recognition models has been improved, it still has some shortcomings. The graph structure in the model has neglecting to incorporate some hidden connections into the model, for example, when people perform the action of "tying shoes", it is difficult for the model to capture the connection between hands and hands, because they have no corresponding connection in the body's natural connections. Also, in previous studies, most of them took the source point 1-neighborhood as the receptive field, and convolved the source node with the node in the region. They did not consider that the nodes in the far region also had influence on the source node. Focusing on the problems, the paper improves the algorithm from the following two aspects:

- Firstly, the static human body structure is connected to make it become a dynamic connection mode, and the corresponding connection is increased through learning between two nodes that are not connected by themselves.
- Secondly, the filter in the image convolution is processed through the Inception structure to increase the impact information of each joint source point.

On the basis of these two aspects, an action recognition model of multi-filter dynamic graph convolutional neural network is proposed. Finally, an action model recognition study was carried out on the Kinetics-keleton dataset, and the results verified the recognition accuracy of the multi-filter dynamic graph convolutional neural network model.

## 2. Spatio-temporal modeling of skeleton sequences

Skeleton graph is a kind of topological graph, which has attracted much attention in the field of pose estimation and action recognition. The nodes in the figure are composed of two or three dimensional coordinates of the human joints, and the edges represent the natural connectivity between the joints. The skeleton diagram sequence is composed of the skeleton diagram corresponding to each frame of action in the video. Each frame of skeleton diagram contains N nodes, corresponding to important joints in the human body. Take two-dimensional coordinates as an example, the node can be expressed as vector $v=(x, y)$, the joint vectors from the first joint to the Nth joint are expressed as set $V=\{v_i \mid v_1, v_2, \ldots, v_N\}$. A skeleton diagram can be expressed as $G=(V,E)$, where $V$ is the set of nodes, E is the edge connected between nodes, the formula is $E=\{v_i v_j \mid v_i \in V, v_j \in V\}$. A skeleton sequence contains T skeleton diagrams, so it can be represented as $Seq=\{G_i \mid G_0, G_1, G_2, \ldots, G_T\}$.

Skeleton graph sequence can only represent the node information of each frame, which represents a kind of spatial dimension information, but it can't represent the information on the time dimension, it can't convey the information of node position changes from frame to frame, therefore, modeling the framework diagram sequence in the time and spatio, connecting the same node between two adjacent frames to obtain the spatio-temporal model. For example, see Fig.1.

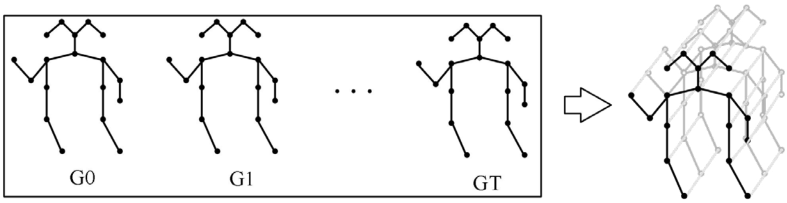


Fig. 1. Spatio-temporal modeling of skeleton sequences.

## 3. Multi-filter dynamic graph convolution network

### *3.1. Network architecture*

The network model consists of a total of 9 layers of space-time residual blocks, each space-time residual block is composed of multi-filter dynamic graph convolution network, temporal convolutional network and common residual structure. The window size of temporal convolutional network is 9, and the residual structure uses dropout with probability 0.5 to prevent overfitting. When the data just enters the model, it passes through a BN layer, which is used to normalize the input data. A global pooling layer is used at the end to map the characteristics of different samples into same size, and finally provided to the softmax classifier for classification prediction of actions. The network model of action recognition is shown in the Fig. 2.

Fig. 2. Network model for action recognition.

### 3.2. Multi-filter graph convolution structure

In the study of skeleton behavior characteristics based on graph convolution, considering the characteristics of motion recognition, filter classification can improve the effect of feature extraction. literature [1] uses the strategy of graph division to divide filters into three categories, which respectively represent centripetal motion, centrifugal motion and static motion. Such a division is considered reasonable by comparison with other divisions, however, in previous studies, the source point 1- neighbourhood in the field has been used as the filter, it only applies the graph convolution with constant filter size to the node-based classification task, but does not solve the problem of graph heterogeneity, it does not take into account the possible effects of distant nodes on it. Therefore, the paper proposes a network structure similar to Inception -- multi-filter structure, it uses filters of different dimensions to convolve the nodes and finally fuses the feature map together.

Inception Structure first appeared in GoogLeNet proposed in literature [9], which is a scheme to improve network performance and act on a convolutional neural network. Prior to Inception, the most straightforward solution to improve network performance was to increase the number of layers or channels per layer of the network, but such an approach would lead to overfitting problems and would significantly increase computation. Therefore, Inception may cluster sparse matrices into relatively dense submatrices to preserve filter level sparsity. The approach of the Inception structure is to use convolution kernels of different sizes to extract features for final splicing, which means the fusion of features of different scales.

The article extends the Inception structure to graph convolution in conjunction with a zoning strategy. 1-neighbourhood filter (k1) and 2- neighbourhood filter (k2) were used to extract the features of skeleton diagrams. The nodes in each neighborhood follow the partitioning strategy, which divides the convolution kernel into three categories: root node, centripetal set (containing nodes closer than the root node and at the same distance as the root node) and centrifugal set, then connect the feature map together. This method solves the problem the influence of distant nodes on source nodes cannot be obtained. The convolution structure of multi-filter graph is shown in Fig.3.

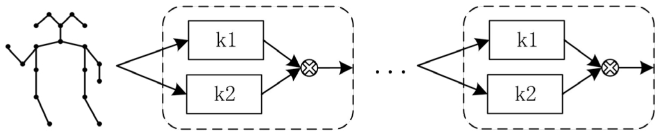


Fig. 3. The convolution structure of multi-filter graph.

### 3.3. Dynamic skeleton diagram

In the research of action recognition based on skeleton graph, most of the researches adopt static graph, namely the physical connection of skeleton, which is a graph structure set according to prior knowledge, and the structure is fixed in all layers. literature [1] proposes an attention mechanism. However, the method only focuses on the strong and weak connections between nodes. For some nodes that are closely related in the movement, it cannot produce connections that do not exist between nodes. Therefore, a new attention mechanism is proposed in this paper, which can turn the static skeleton graph into a dynamic graph structure. It can generate new connections between physically disconnected nodes through learning, making the motion recognition algorithm more reasonable and accurate.

The original formula of convolution of spatial graph is:

$$f_{out}(v_i) = \sum_{v_j \in \beta_i} \frac{1}{Z_{ij}} f_{in}(v_j) \cdot w(l_i(v_j)) \tag{1}$$

Where, $f$ is the feature mapping relation, $f_{in}$ is the input feature map, $f_{out}$ is the output feature map, $v$ is the node in the figure, which is the sampling area of $v_i$ convolution, $w$ is the weight function, which provides a weight vector according to each input, and $v_i$ is a mapping function, which represents the partitioning strategy of each node.

The formula can be converted as follows:

$$f_{out} = \sum_{k}^{K_v} W_k(f_{in}A_k) \odot M_k \tag{2}$$

In the transformed formula, $K_v$ is on behalf of the kernel size of the space dimension. $K_v$ is 3 known by the partitioning strategy, and $M_k$ is an attention mechanism, and its relationship with the matrix $A_k$ is the dot product.

The principle of dynamic skeleton diagram is also to introduce an attention mechanism:

$$f_{out} = \sum_{k}^{K_v} W_k f_{in}(A_k + M_k) \tag{3}$$

Among them, $M_k$ and matrix $A_k$ are the relations of addition and sum, which can not only dynamically change the strength relationship between joint connections, but also generate new connections that do not exist in the original topology diagram, making it optimized together with other network parameters, which increases the flexibility of the action recognition model.

## 4. Experiment and result analysis

### *4.1. Experiment dataset*

Kinetics data set is the largest unconstrained action recognition data set at present, which contains more than 240000 video clips and covers up to 400 types of human actions. The video content includes human-object interaction and human-human interaction, ranging from daily human behavior and sports to complex interactive actions, with each video clip lasting about 10 seconds. Due to the Kinetics data set only provide RGB video clips, and no skeleton data, so this article with the help of Openpose pose estimation algorithm, each frame of video joint position estimation, had 18 points and node degree of confidence, each vector coordinates (x, y, acc), average and selection on each frame joints, record their joint coordinates and confidence as a feature. In this paper, the Kinetics data set processed by Openpose is called Kinetics-skeleton data set.

### *4.2. The experimental configuration*

In the experiment, the reproducible filling method is used to ensure that each sequence data is within 300 frames. The experiment was developed based on Pytorch deep learning framework. The optimization strategy adopted random gradient descent SGD. The initial learning rate was set as 0.1, and the learning rate decreased to 0.0001 after 50 rounds of training, and the training batch size was 64.

### *4.3. Analysis of experimental results*

The recognition performance is evaluated by the accuracy rate of top-1 and top-5. The training set of the data set contains 240000 videos, while the test set contains 20000 videos.

Table 1 is the comparison between the action recognition algorithm in this paper (MFD-GCN) and other classical algorithms. It can be seen that the accuracy of the algorithm in this paper (MFD-GCN) is improved by 2.1% and 2% respectively by top-1 and top-5.

Table 1.Comparison between MFD-GCN and other algorithms on Kinetics-skeleton dataset.

| Classical network model | Top-1 accuracy rate(%) | Top-5 accuracy rate(%) |
|---|---|---|
| Deep-LSTM | 16.4 | 35.3 |
| TCN | 20.3 | 40.0 |
| ST-GCN | 30.7 | 52.8 |
| MFD-GCN(ours) | 32.8 | 54.8 |

Loss and accuracy change with the number of training frontal rounds. The fig.4 shows the change curve of Loss and accuracy recorded once every 5 rounds, after 50 rounds of iteration, loss decreased from 5.18 to 2.51. The accuracy rate of top-1 reached 32.8%, and that of top-5 reached 54.8%.

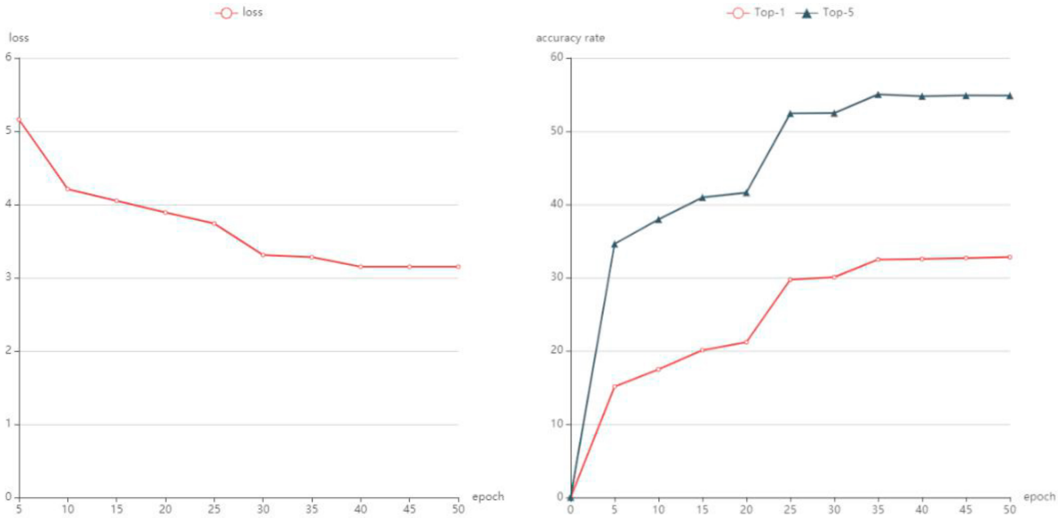


Fig. 4. (a) loss change curve;(b)accuracy rate change curve.

## 5. Conclusion

In this paper, a multi-filter dynamic graph convolutional neural network is proposed to obtain the influence of distant nodes on the origin by increasing the neighborhood range of source points, and to adapt the connection relationship between nodes with different behaviors to the dynamic skeleton graph, thus changing the original feature extraction scheme of skeleton graph.

Tests on Kinetics-skeleton data set extracted from the skeleton showed that the accuracy of top-1 and top-5 of the model reached 32.8% and 54.8% respectively. Compared with the classical model, the model achieved better accuracy and verified the action recognition effect and ability of the model.

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## References

1. S. Yan, Y. Xiong, and D. Lin. Spatial temporal graph convolutional networks for skeleton-based action recognition. In AAAI Conference on Artificial Intelligence, pages 7444–7452, Feb 2018.
2. Kalpit Thakkar and P J Narayanan. Part-based graph convolutional network for action recognition. In BMVC, 2018.
3. Lei Shi, Yifan Zhang. Two-Stream Adaptive Graph Convolutional Networks for Skeleton-Based Action Recognition. In CVPR, 2019.
4. T. N. Kipf and M. Welling. Semi-supervised classifification with graph convolutional networks. In ICLR, Apr 2017.
5. Tang, Y.; Tian, Y.; Lu, J.; Li, P.; and Zhou, J. 2018. Deep progressive reinforcement learning for skeleton-based action recognition. In Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition, 5323–5332.
6. Liangliang Wang, Lianzheng Ge, Ruifeng Li. Three-stream convolutional neural networks. Patten Recognition Letters, 2017.
7. Bilen Hakan, Fernando Basura, Gavves EfstratiosAction. Recongnition with dynamic image networks. In CVPR, 2017.

8. Cao, Z.; Simon, T.; Wei, S.-E.; and Sheikh, Y. 2017b. Realtime multi-person 2d pose estimation using part affifinity fifields. In CVPR.
9. C. Szegedy, W. Liu, Y. Jia, P. Sermanet. Going deeper with convolutions. In CVPR, 2015.
10. Qiuhong Ke, Mohammed Bennamoun, Senjian An, Ferdous Ahmed Sohel, and Farid Boussad. A New Representation of Skeleton Sequences for 3d Action Recognition. 2017 IEEE Conference on Computer Vision and Pattern Recognition (CVPR), pages 4570–4579, 2017. 1, 2, 8.
11. Zhu, W.; Lan, C.; Xing, J.; Zeng, W.; Li, Y.;Shen, L.; Xie, X.; et al. 2016. Co-occurrence feature learning for skeleton based action recognition using regularized deep lstm networks. In AAAI.
12. A. Shahroudy, J. Liu, T.-T. Ng, and G. Wang. Ntu rgb+d: A large scale dataset for 3d human activity analysis. In The IEEE Conference on Computer Vision and Pattern Recognition (CVPR), pages 1010–1019, June 2016.